College of Science and Mathematics

Can Exercise Increase Test Scores and Decrease Stress Levels for Kennesaw State University Students?

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Introduction

Stress is a key component that most students feel daily. With this stress that students feel it can also have a negative effect on their mental health. Implementing exercise as a coping method can help students especially when they are feeling overwhelmed. Studies have shown exercise to be beneficial to students' memory and cognitive function especially since it can be found as a break from students' studies and stress. Using exercise can help students' mental health and can improve people's intelligence levels which can be helpful as college student when they are having trouble focusing on their studies. Therefore, we hypothesize, if one implements exercise as part of your routine as a college student then it could decrease the amount of stress one would feel and increase test scores.

Methods

In order to conduct the research, 548 undergraduate students were surveyed in various metrics, including exercising, mental health, and stress levels. The data obtained was later compared with each individual average test scores in science and mathematics courses. For this study, the individual's average test score was used as the dependent variable while the independent variable used were the average stress levels and hours spend exercising. Using the information obtained from the survey, we analyzed the all the data in an excel spreadsheet to create graphs and tables.

Survey students ⇨ Get average of data ⇨ Create graphs in excel

Results



Figure 1 Graph of Average test scores. Demonstrates the average current course score for Biology 1107, Math 1113, and Math 1111.

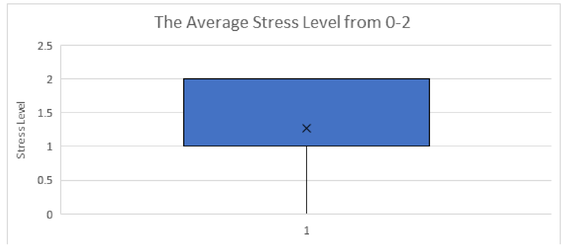


Figure 2 Graph of Average stress level. Shows the median of how stressed students feel about studying for exams, preparing for classes, and homework. The "X" on the graph shows the median

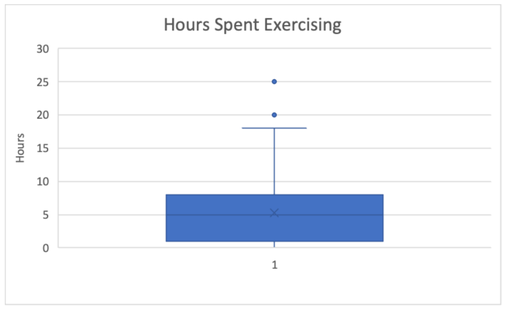


Figure 3 Graph of Average time spent exercising. Shows the median amount of time spent exercising per week. The "X" on the graph shows the median

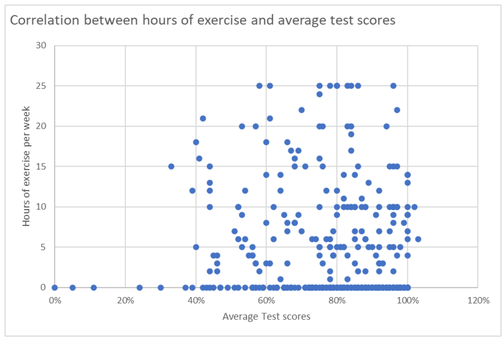


Figure 4 Scatter Graph of Hours of exercise V. Test scores. Demonstrates the correlation between the hours of exercise per week (y-axis) and the average test scores that each participant obtained (x-axis).

- Shown in figure 1, The majority of students have test scores within the range of 60% to 90%
- Shown in figure 2, Most students report a higher stress level when preparing for classes and exams
- Shown in figure 3, Most students spend from 1 to 7 hours exercising during the week
- Shown in figure 4, The correlation shows that students that spend time to exercise also have on average higher grades

Discussion

- Students reported high stress levels due to heavy workload and time spent on assignments and exams.
- Figure 2 showed that science and math courses contributed significantly to stress levels.
- Hypothesis that exercise could reduce stress and improve test scores was not supported by Figure 4. Indicated no correlation between gym time and test scores.
- Despite exercising being thought to alleviate stress, survey results did not support the expected positive impact.
- Using the different figures and tables that represented: the biology 1107 current course score (Figure 1), median amount of time spent at the gym (Figure 3) and provided stress levels (Figure 2) it can be concluded that exercise does not necessarily affect stress levels or performance in the classroom.
- Acknowledging limitations, such as sample size and survey design is crucial for interpretation.
- Overall, our study did not find conclusive evidence supporting the hypothesis of exercise positively influencing academic performance or reducing general stress.

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